



Near Infrared Spectroscopy (NIRS) for Agronomy

Martin Ecartot, INRAE / AGAP Institut





Map of the course

- Phenotyping and NIRS
- NIRS data : how spectrum is generated.
- Applications in agronomy.
- Preprocessing and Principal Component Analysis.
- Let's work ! Application with 
 - Handling data
 - Preprocessings
 - PCA
- Discussion

NIRS and Phenotyping

A need for phenotyping for **breeders** and product quality.

- Human eye

- Quality of ears/grains
- Leaves characterisation (senescence, diseases)
- Plant Architecture (Height, leaf angle,...)
- Set-up Evaluation



- Chemical analysis

- Nitrogen content (leaf, whole plant)
- Grain Proteins
- Lipids

- Genotyping

- Genomique Selection

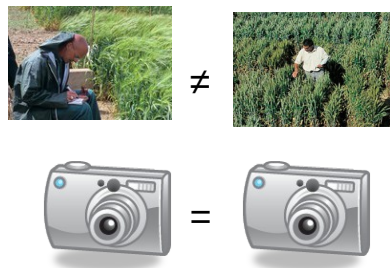


NIRS and Phenotyping

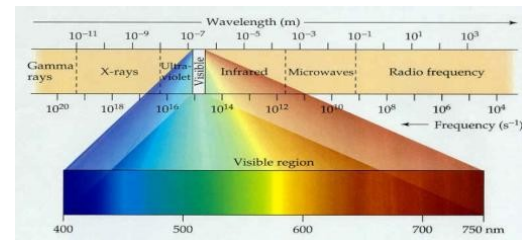
Optical or « High-Throughput » Phenotyping

Advantages

- Objective and repeatable measurements
- Low cost
- No chemicals
- Non-destructive measurements
- Fast
- Easy to transport
- Infrared vision



Wheat grains
measured with
NIRS



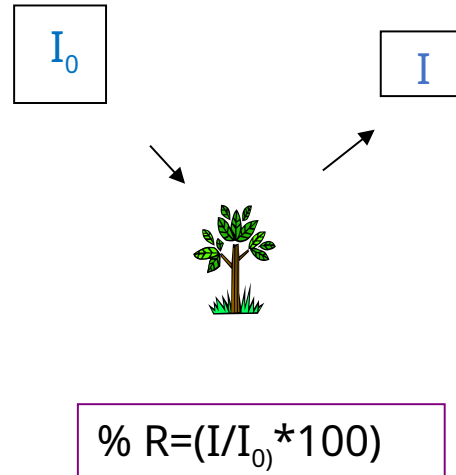
Disadvantages

- Sensors must be adapted to the measured product
- Calibration required

NIRS and Phenotyping

Why Near-Infrared?

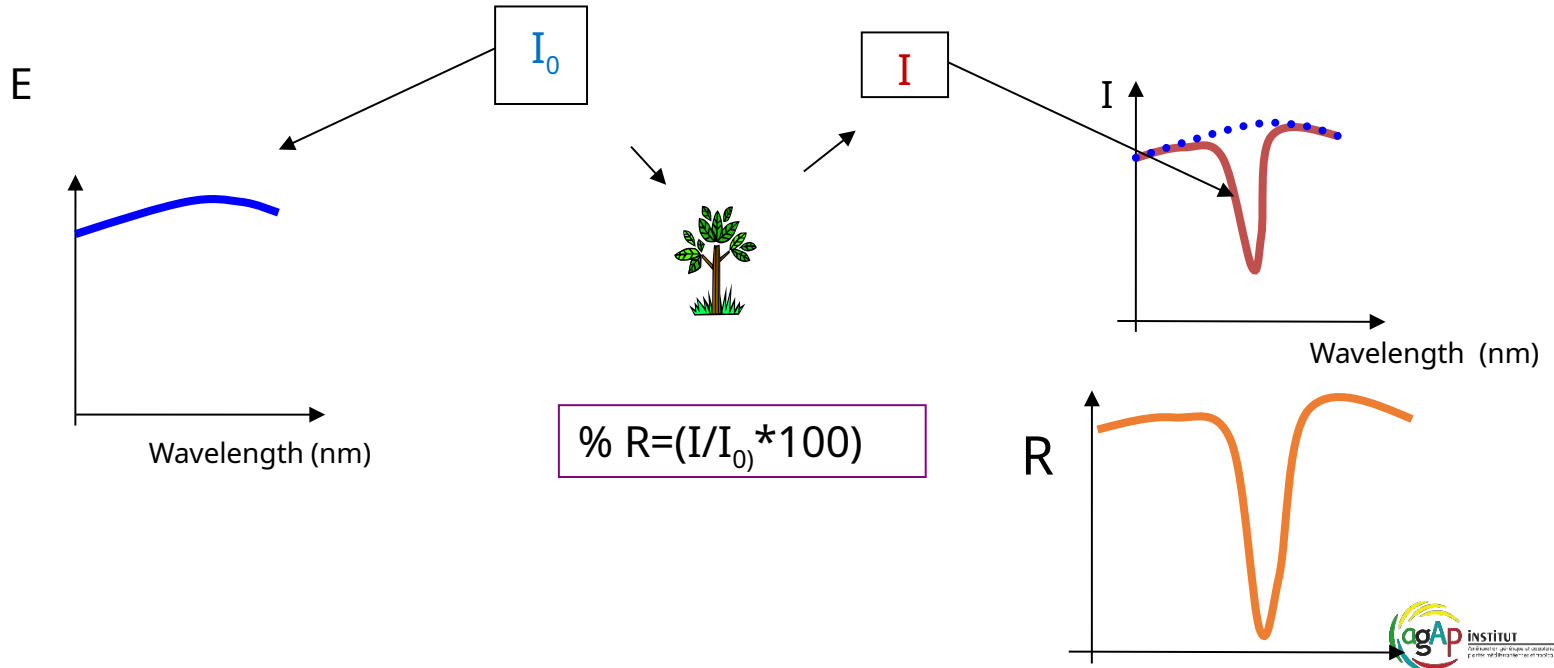
Optical Measurement



NIRS and Phenotyping

Why Near-Infrared?

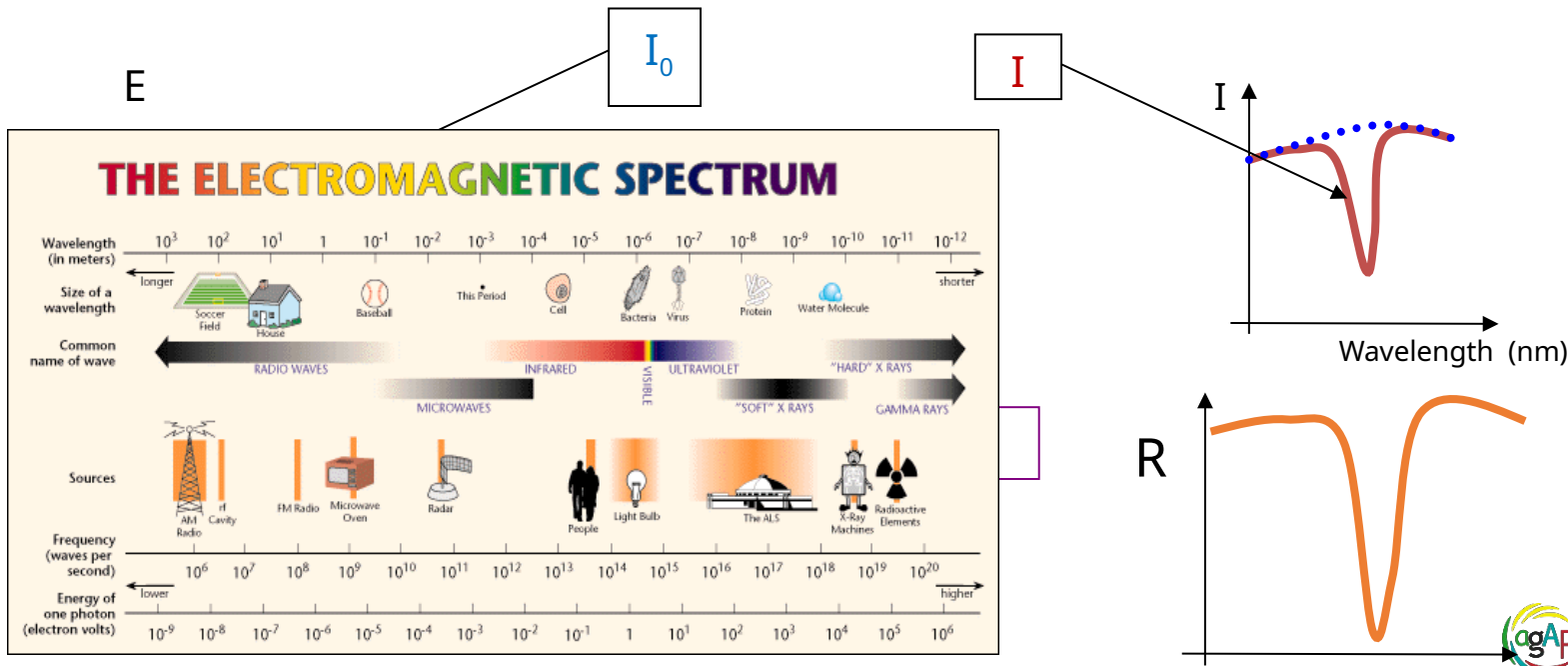
Spectroscopic Measurement



NIRS and Phenotyping

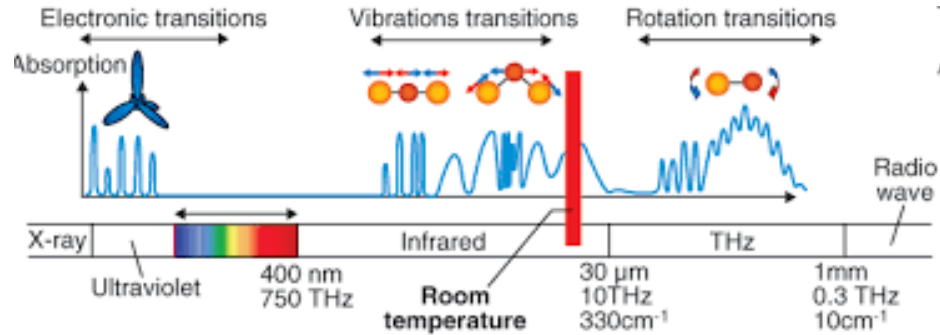
Why Near-Infrared?

Spectroscopic Measurement



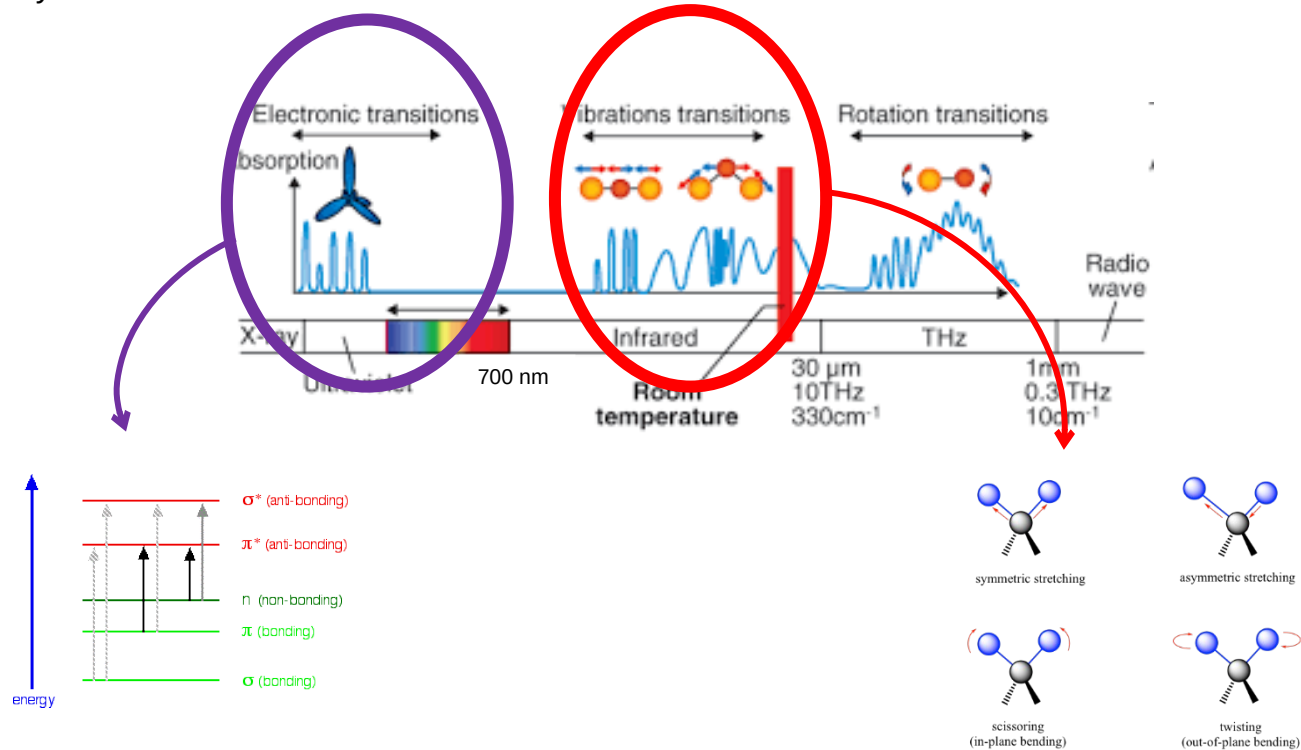
NIRS and Phenotyping

Why Near-Infrared?



NIRS and Phenotyping

Why Near-Infrared?

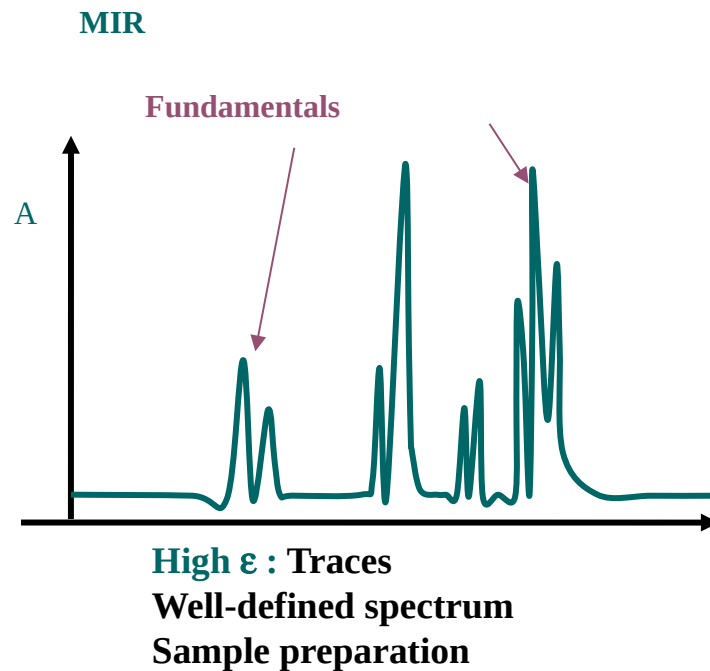
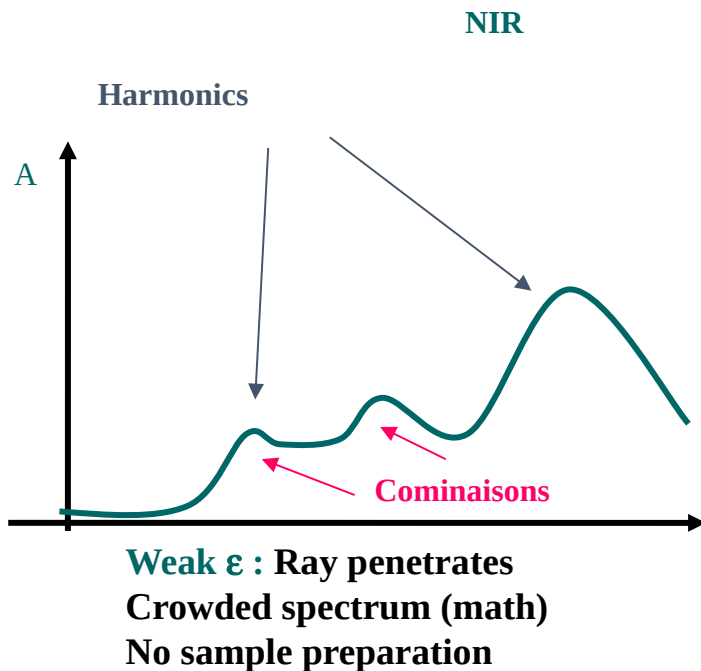


In the near-infrared,
Only harmonic oscillations and combinations
of fundamental MIR oscillations are measured.



NIRS and Phenotyping

Why Near-Infrared?



NIRS and Phenotyping

Why Near-Infrared?

Identify and quantify a compound:

Beer-Lambert law

$$A = \epsilon \cdot l \cdot c = -\log(R)$$

ϵ : absorption coefficient

l : optical path length

c : compound concentration

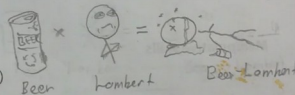
Beer-Lambert law states that this absorbance is directly proportional to the concentration of solution.

$$A = \epsilon \cdot c \cdot l$$

where c = concentration of solution

l = length of light path (1 cm)

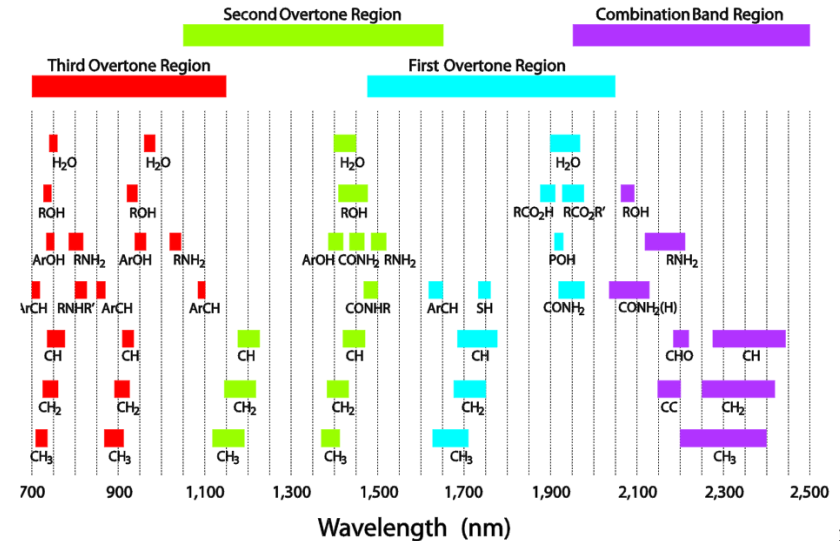
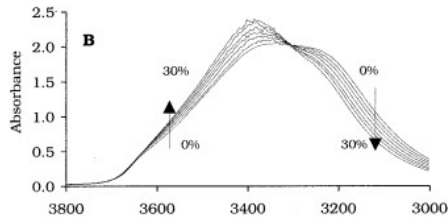
ϵ = extinction coefficient



NIRS and Phenotyping

Why Near-Infrared?

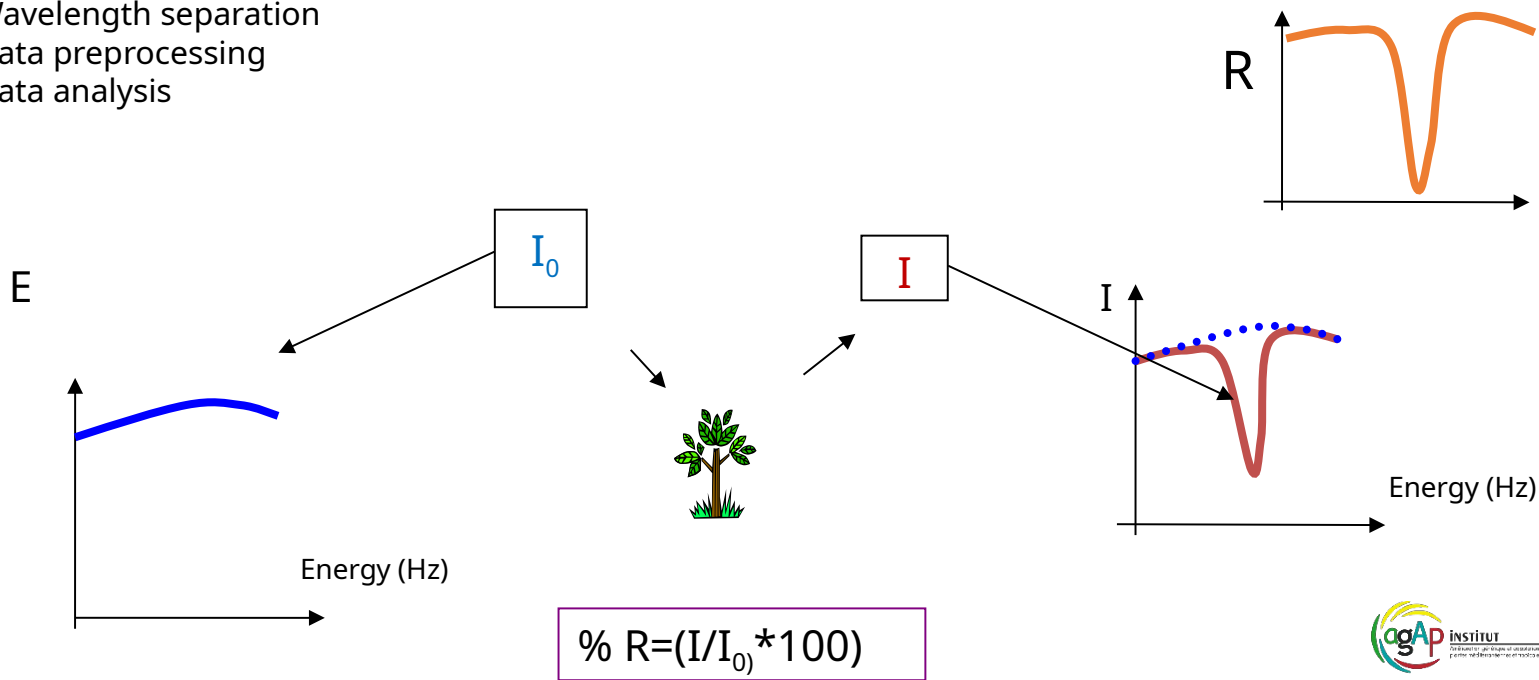
- Advantages
 - Deeper light penetration than MIR (>1 mm)
 - Many sensitive compounds in this range
- Disadvantages
 - Broad bands → Overlap
 - Low sensitivity (difficult if < 0.1%)
 - Numerous interferences
 - Temperature
 - Particle size
 - Interactions



NIRS: Data generation and processing

NIRS data, from A to Z

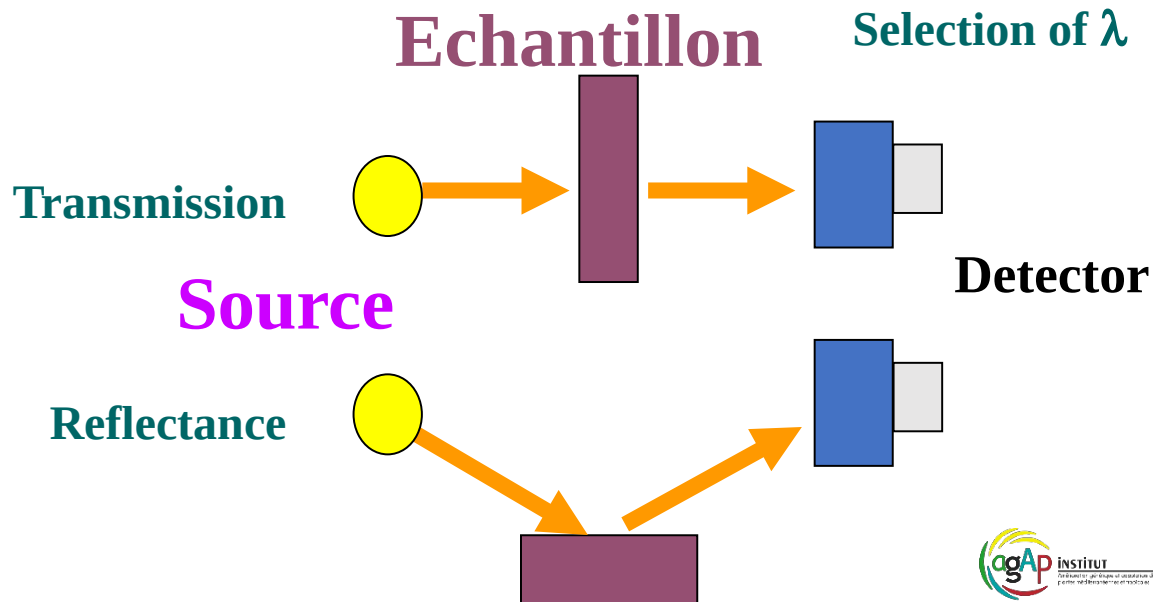
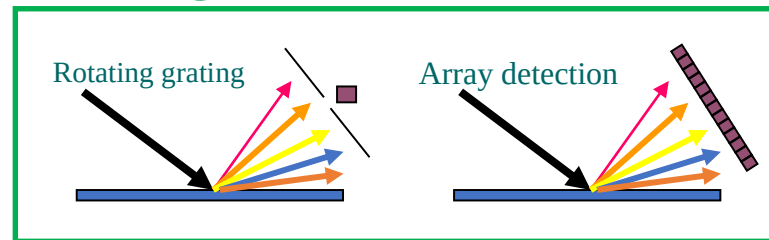
- Sample presentation
- Wavelength separation
- Data preprocessing
- Data analysis



NIRS: Data generation and processing

NIRS data, from A to Z

- Sample presentation
 - Reflection/Transmission
- Wavelength separation
 - Filter
 - Diffraction grating
 - Fourier transform
- Data preprocessing
- Data analysis



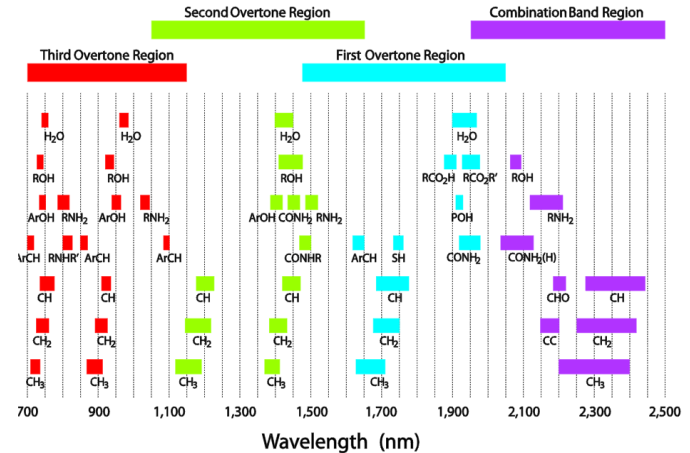
NIRS: Data generation and processing

NIRS data, from A to Z

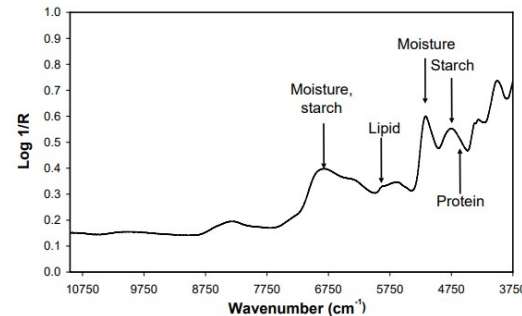
- Data preprocessing

What is a NIR spectrum?

- Light/matter interaction
- Influence of chemical composition
- Influence of physics
- Disruptions: electronics, contamination,...



Wheat Grain (reflectance)



Wheat flour, Bhandari et al, 2006

NIRS: Data generation and processing

NIRS data, from A to Z

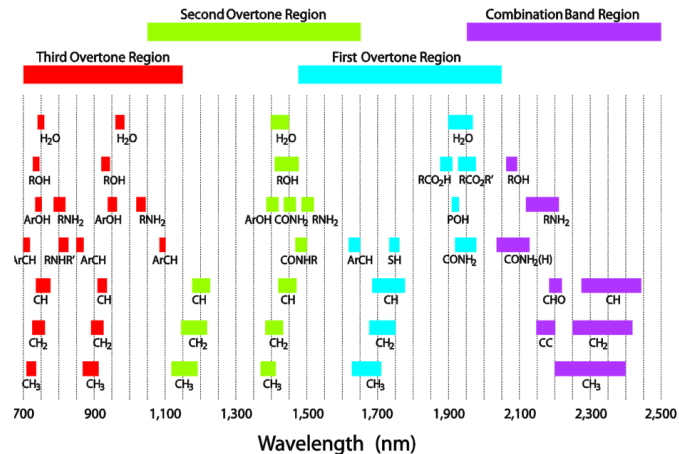
- Data preprocessing

What is a NIR spectrum?

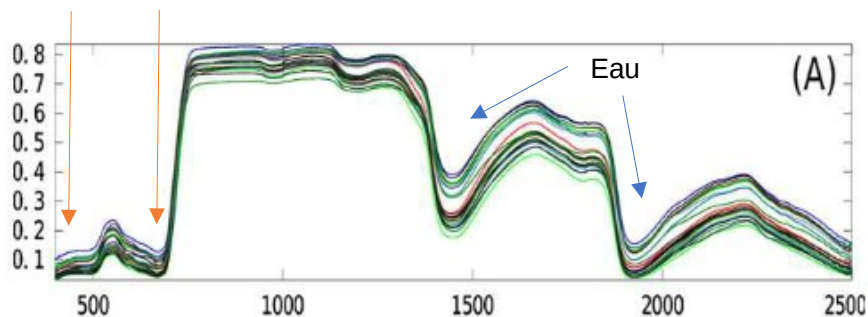
- Light/matter interaction
- Influence of chemical composition
- Influence of physics
- Disruptions: electronics, contamination,...



Leaf Spectra



Chlorophylls



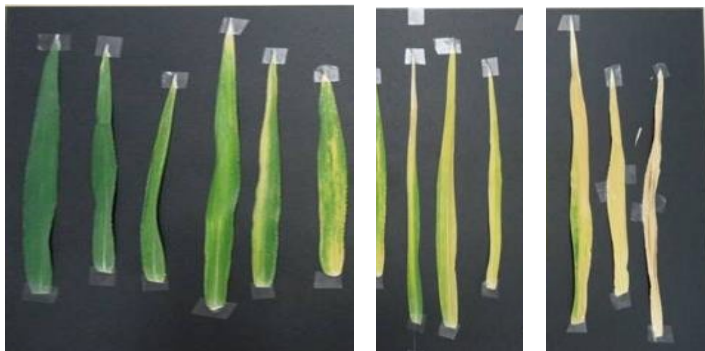
NIRS: Data generation and processing

NIRS data, from A to Z

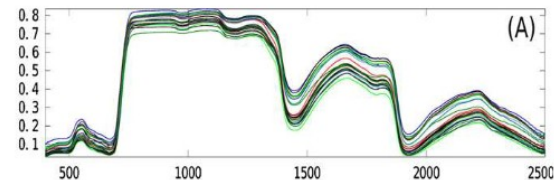
- Data preprocessing

What is a NIR spectrum?

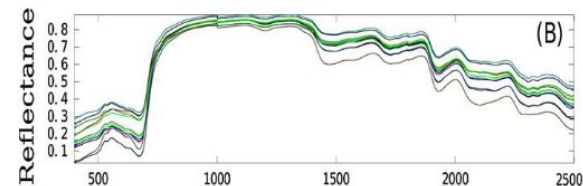
- Light/matter interaction
- Influence of chemical composition
- Influence of physics
- Disruptions: electronics, contamination,...



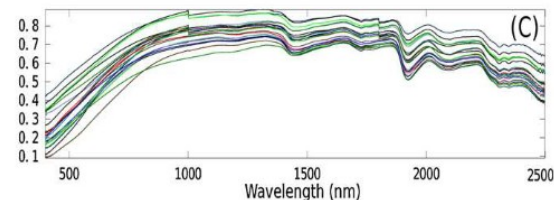
Fresh leaves



Dried leaves



Senescent leaves



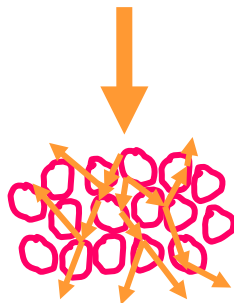
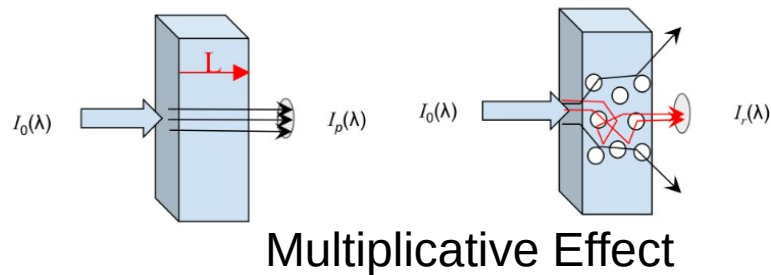
NIRS: Data generation and processing

NIRS data, from A to Z

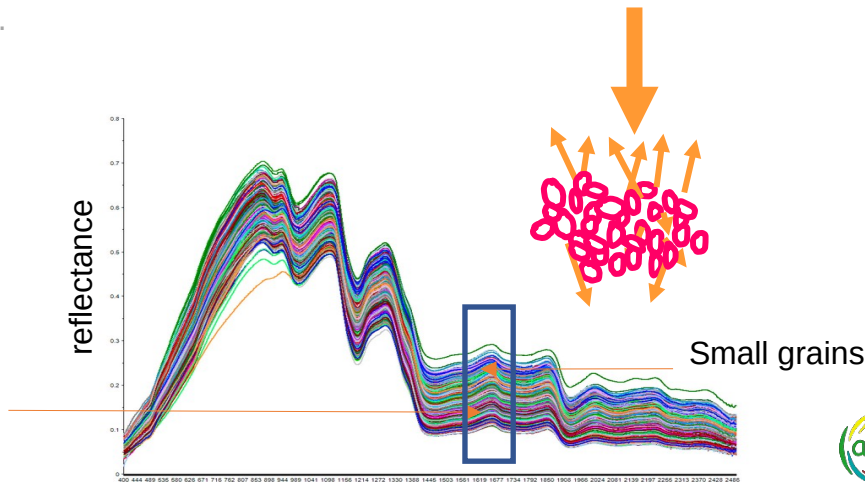
- Data preprocessing

What is a NIR spectrum?

- Light/matter interaction
- Influence of chemical composition
- Influence of physics
- Disruptions: electronics, contamination,...



Big Grains



Small grains

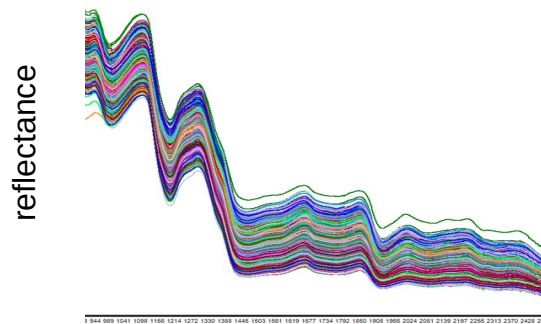
NIRS: Data generation and processing

NIRS data, from A to Z

- Data preprocessing

What is a NIR spectrum?

- Light/matter interaction
- Influence of chemical composition
- Influence of physics
- Disruptions: electronics, contamination,...



Base line drift:
The scattering of light increases with
wavelength

NIRS: Data generation and processing

NIRS data, from A to Z

- Data preprocessing

What is a NIR spectrum?

- Light/matter interaction
- Influence of chemical composition
- Influence of physics
- Disruptions: electronics, contamination,...

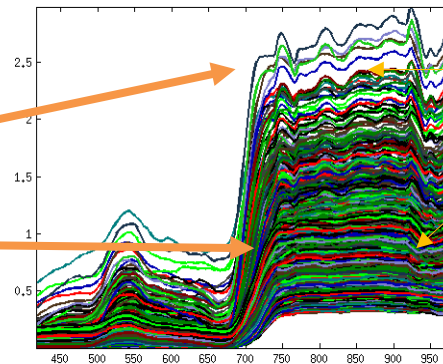
Specular Reflexion



(a) comportement spéculaire



(b) comportement diffus



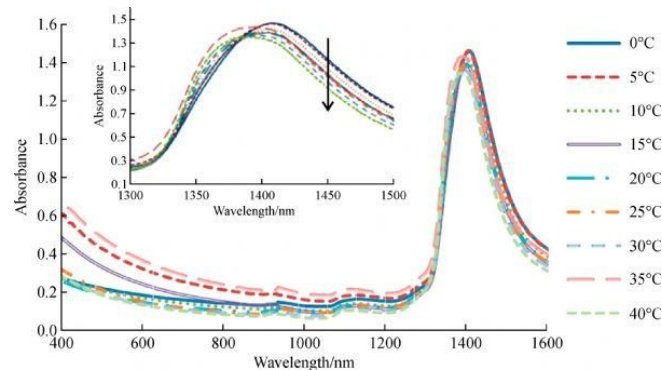
NIRS: Data generation and processing

NIRS data, from A to Z

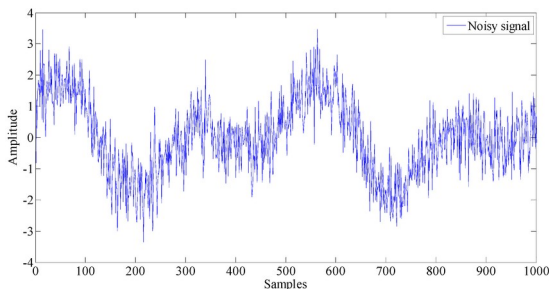
- Data preprocessing

What is a NIR spectrum?

- Light/matter interaction
- Influence of chemical composition
- Influence of physics
- Disruptions: electronics, contamination,...



Xu et al, 2019





NIRS: Data generation and processing

NIRS data, from A to Z

- Data preprocessing

What is a NIR spectrum?

- Light/matter interaction
- Influence of chemical composition
- Influence of physics
- Disruptions: electronics, contamination,...

The mathematical processes that allow us to take into account its various sources of influence are called **preprocessing**.

We'll talk about that right after...

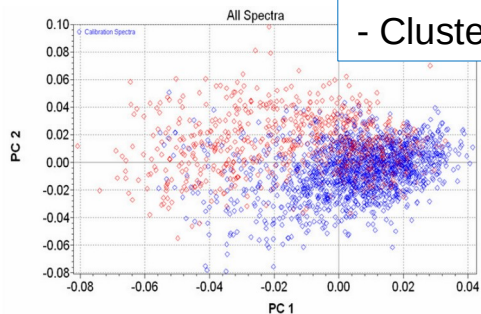
NIRS: Data generation and processing

NIRS data, from A to Z

- Sample presentation
- Wavelength separation
- Data preprocessing
- Data analysis

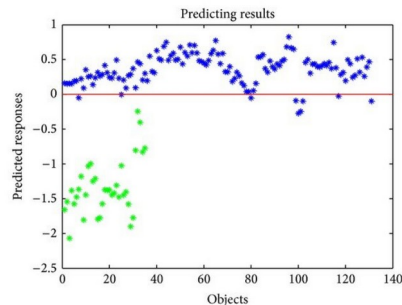
Exploratory methods

- PCA
- Clustering



Predictive methods

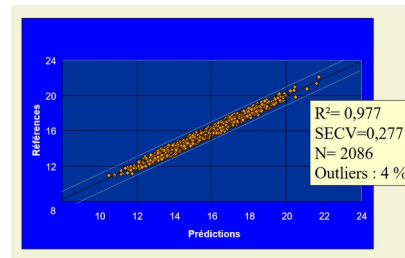
Qualitative models



Yan et al, 2014

Quantitative models

$$\begin{matrix} \text{réponses} & & \text{prédicteurs} & & \text{b-coefficients} & & \text{résidus} \\ \left\{ \begin{matrix} Y \\ \end{matrix} \right\} & = & \left\{ \begin{matrix} X \\ \end{matrix} \right\} \cdot \left\{ \begin{matrix} B \\ \end{matrix} \right\} + \left\{ \begin{matrix} R \\ \end{matrix} \right\} \end{matrix}$$





NIRS : Applications in Agronomics

A technique used in many fields

Food processing / Agronomy

Pharmacy

Medicine

Soil science

Geology

Waste sorting

Fraud detection

Forensic science

NIRS : Applications in Agronomics

On grains:

Protein Content (CTPS)



Moisture / Dry matter
Protein
Sugars
Fat

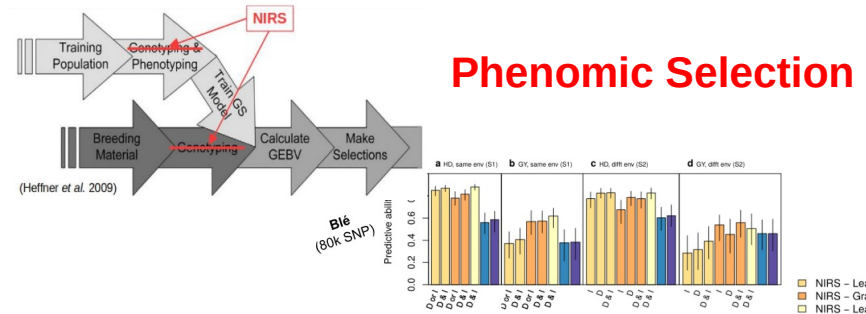


Soja OGM
Roundup Ready™



Soja conventionnel

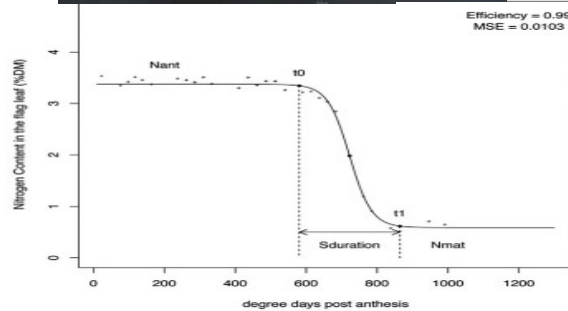
Correct classif.
92,4%



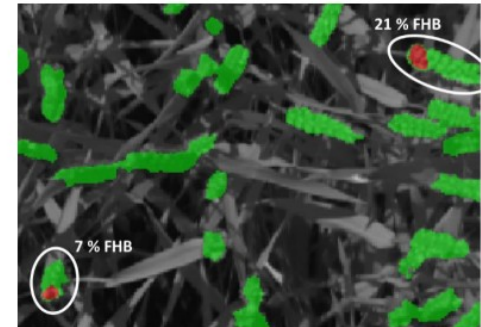
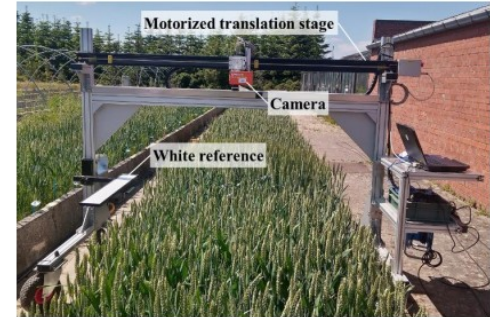
NIRS : Applications in Agronomics

On plants:

Leaves nitrogen content



Fusarium head blight assessment (CRA-W)



NIRS : Applications in Agronomics

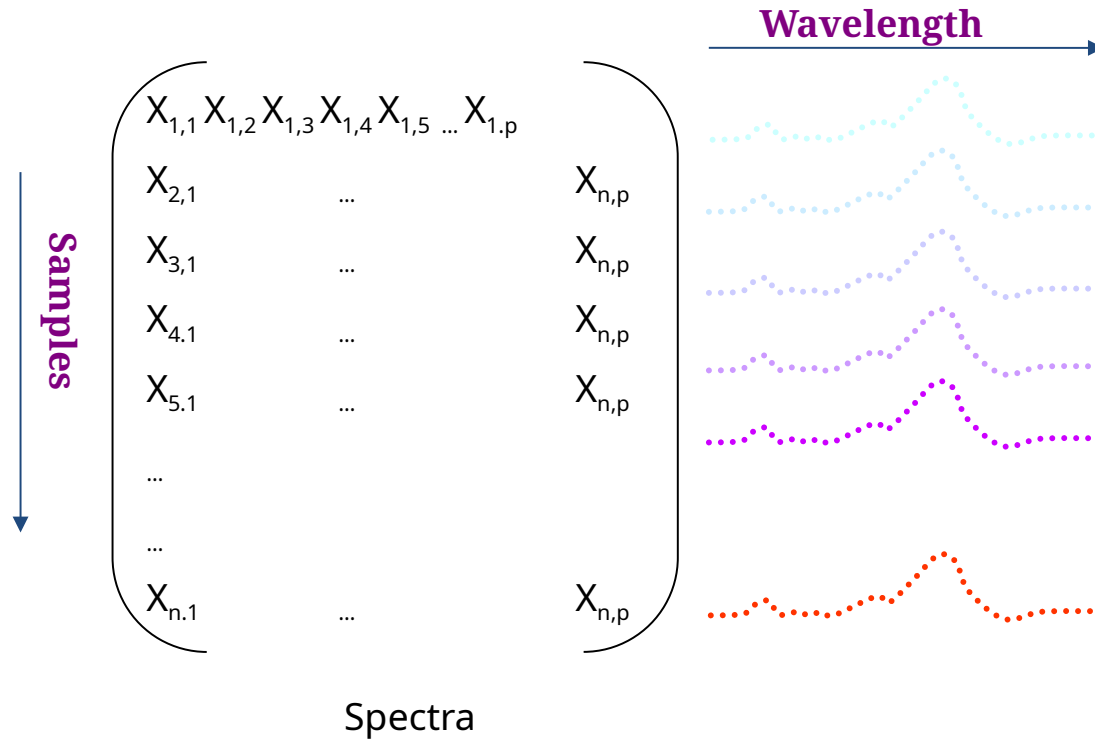
On fruits:

Online analysis of the sugar content of apples



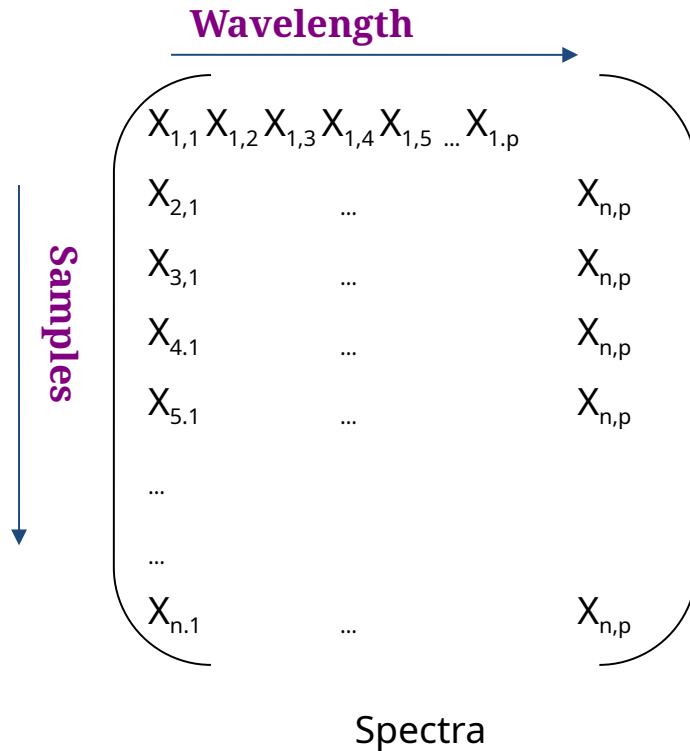
NIRS: What precautions should be taken when processing data?

Enter the Matrix...



NIRS: What precautions should be taken when processing data?

Enter the Matrix...



In the Spectra Matrix:

Rows = Individual

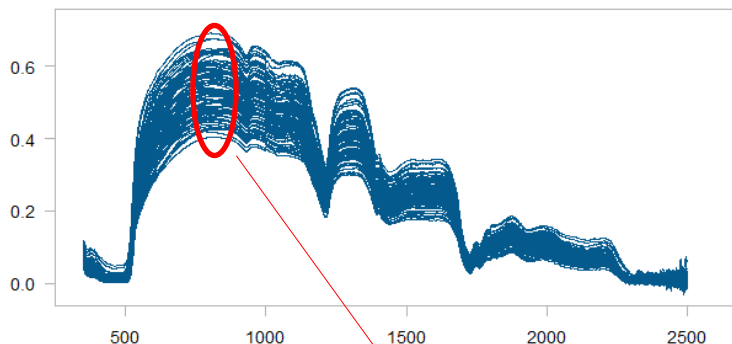
Columns = Variables

Exactly like :

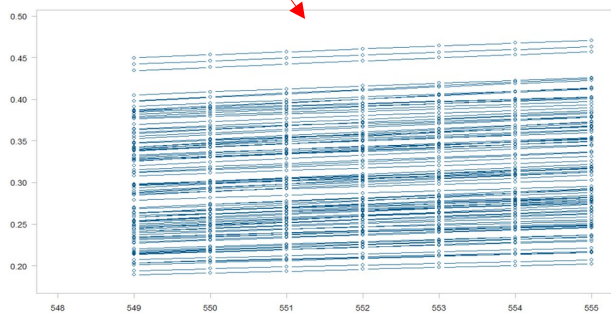
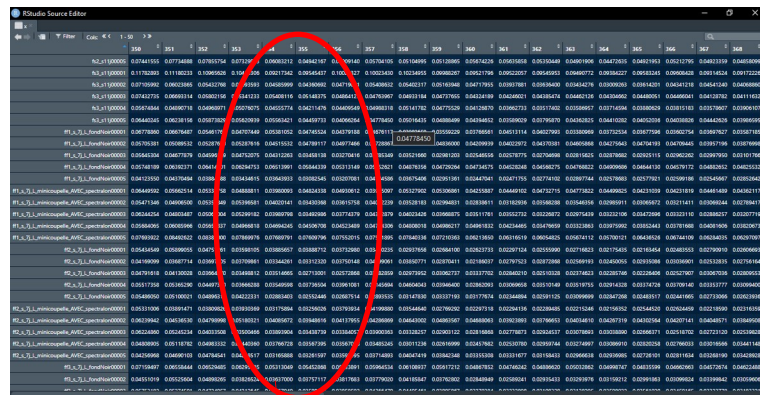
```
> head(iris)
  Sepal.Length Sepal.Width Petal.Length Petal.Width Species
1          5.1         3.5          1.4          0.2  setosa
2          4.9         3.0          1.4          0.2  setosa
3          4.7         3.2          1.3          0.2  setosa
4          4.6         3.1          1.5          0.2  setosa
5          5.0         3.6          1.4          0.2  setosa
6          5.4         3.9          1.7          0.4  setosa
```

NIRS: How to process data

Enter the Matrix...



=



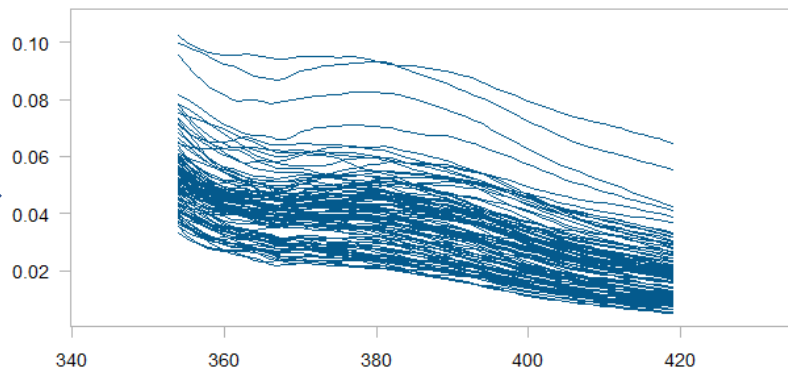
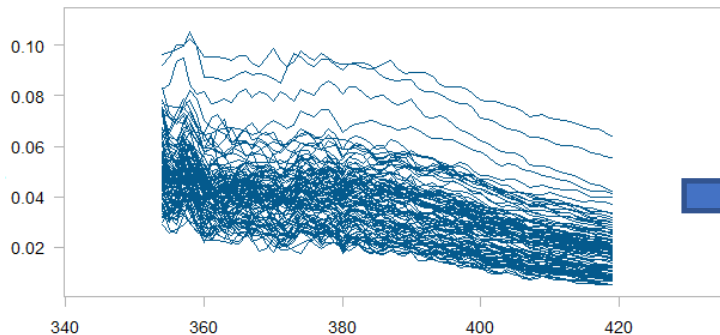
=

	549	550	551	552	553
fs2_s11j00005	0.3977160	0.4022721	0.4065713	0.4107540	0.4148661
fs3_s11j00001	0.4421590	0.4472801	0.4496612	0.4531382	0.4564772
fs3_s11j00002	0.3392124	0.3430121	0.3466758	0.3501895	0.3536320
fs3_s11j00003	0.3639590	0.3665730	0.3690074	0.3713782	0.3737252
fs3_s11j00004	0.3469997	0.3506492	0.3541951	0.3575716	0.3608734
fs3_s11j00005	0.3261270	0.3302771	0.3342712	0.3381723	0.3419623

NIRS: How to process data

Preprocessing techniques to correct disturbances

- Reduce "white" noise
 - Smoothing. e.g., moving average



NIRS: How to process data

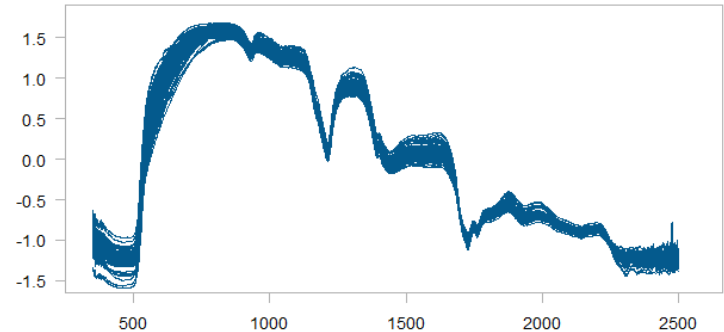
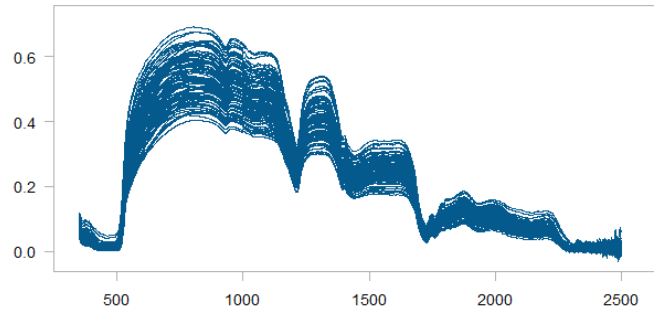
Preprocessing techniques to correct disturbances

- Reducing a multiplicative effect

- SNV Normalisation

For each spectrum, we subtract the mean and divide by the standard deviation.

$$x_{i,j}^{\text{SNV}} = \frac{(x_{i,j} - \bar{x}_i)}{\sqrt{\frac{\sum_{j=1}^p (x_{i,j} - \bar{x}_i)^2}{p-1}}}$$

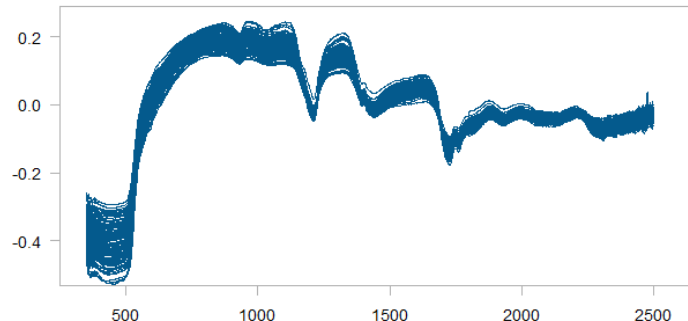
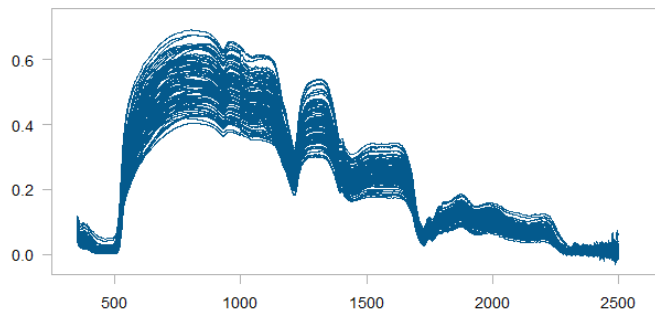


- Multi-scatter correction (MSC)
- Logarithm

NIRS: How to process data

Preprocessing techniques to correct disturbances

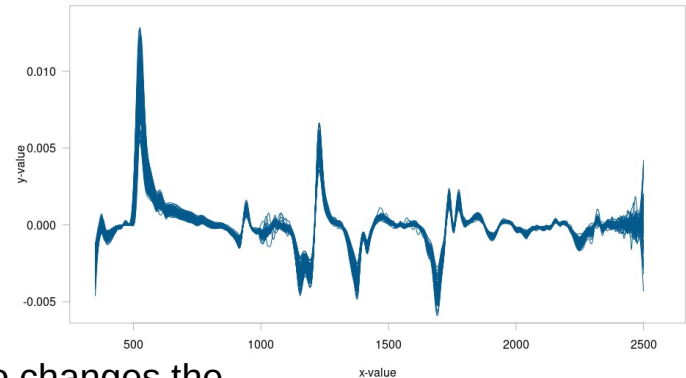
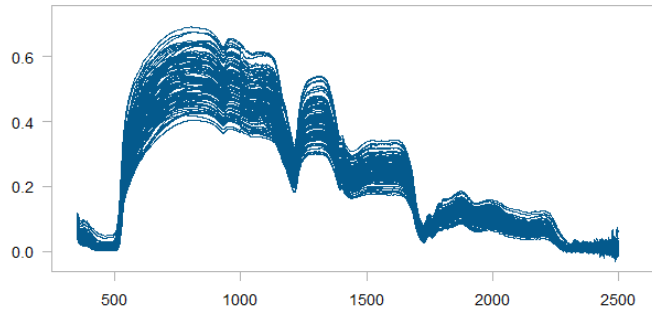
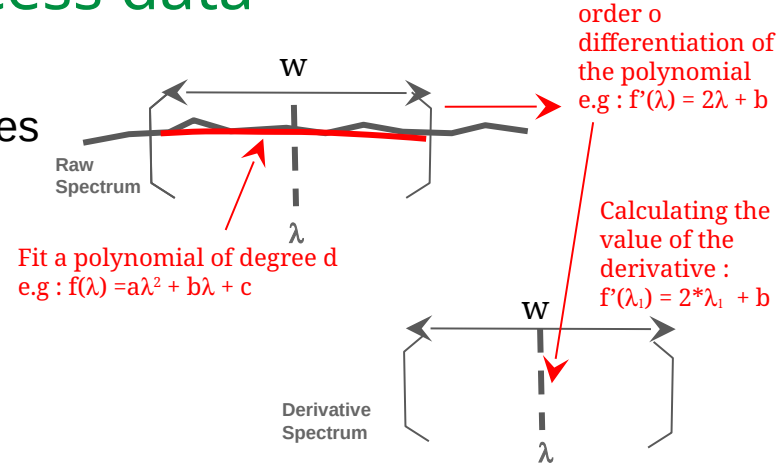
- Reducing baseline drift
 - Algorithm « Detrend »: We subtract a fitted polynomial of order (1, 2, ...)



NIRS: How to process data

Preprocessing techniques to correct disturbances

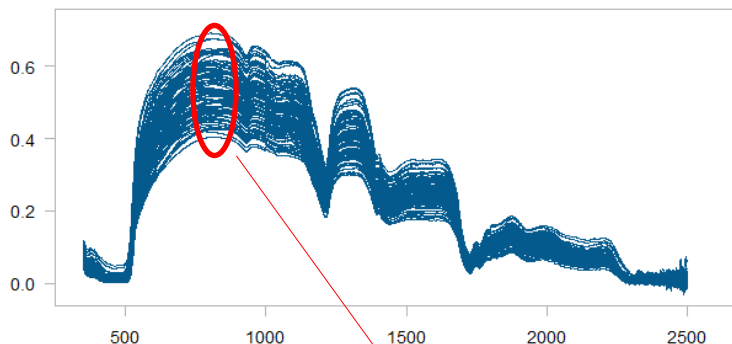
- Reducing baseline drift
 - Derivative (1st, 2nd)
 - Savitzky-Golay Algorithm



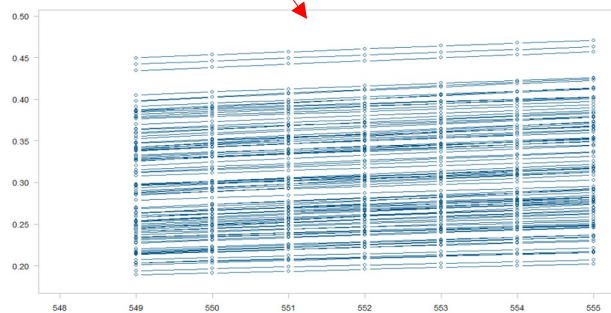
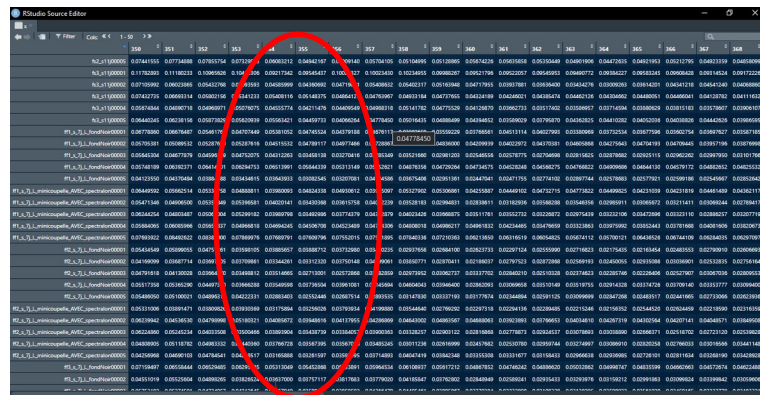
It also changes the resolution of the peaks.!!

NIRS: How to process data

Enter the Matrix...



=



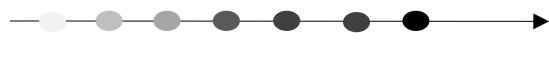
=

	549	550	551	552	553
fs2_s1j00005	0.3977160	0.4022721	0.4065713	0.4107540	0.4148661
fs3_s1j00001	0.4421590	0.4427801	0.4496612	0.4531382	0.4564772
fs3_s1j00002	0.3392124	0.3430121	0.3466758	0.3501895	0.3536320
fs3_s1j00003	0.3639590	0.3665730	0.3690074	0.3713782	0.3737252
fs3_s1j00004	0.3469997	0.3506492	0.3541951	0.3575716	0.3608734
fs3_s1j00005	0.3261270	0.3302771	0.3342712	0.3381723	0.3419623

NIRS: How to process data

How to classify spectra

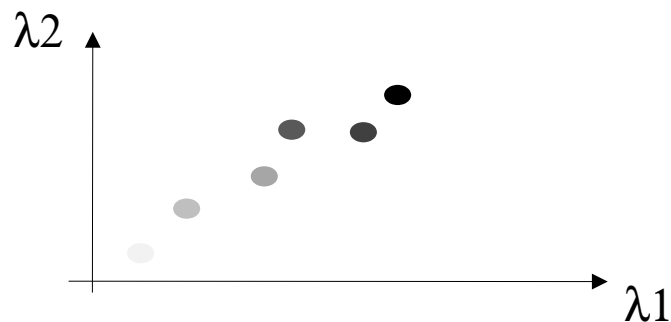
If the spectrometer measures 1 wavelength



● Low content

● High content

If the spectrometer measures 2 wavelengths



NIRS: How to process data

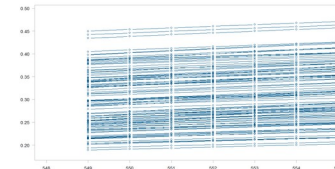
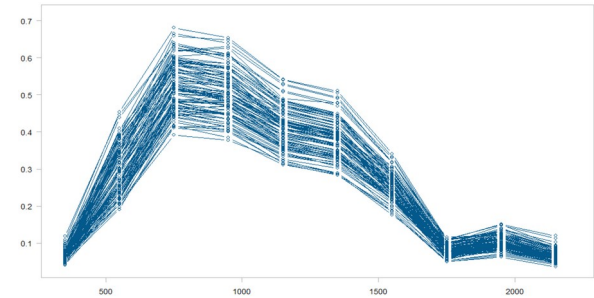
How to classify spectra

If the spectrometer measures 2000 wavelength



Which wavelengths should I choose?

- If we choose 1 wavelength every X...
 - The peaks are no longer visible.
- If we choose a limited range
 - Very high collinearity between variables





NIRS: How to process data

How to classify spectra

If the spectrometer measures 2000 wavelength

Solution?

.....



NIRS: How to process data

How to classify spectra

If the spectrometer measures 2000 wavelength

Which wavelengths should I choose?

Principal Component Analysis!

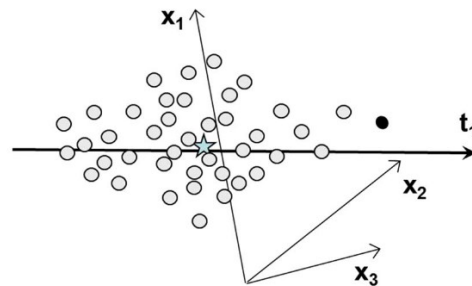
NIRS: How to process data

Principal Component Analysis

- To explore the data (individual ranking, outliers)
- Determine the importance of wavelengths

A principal component = a linear combination of wavelengths

$$CP1 = \alpha \times \lambda_1 + \beta \times \lambda_2 + \dots + \gamma \times \lambda_p$$

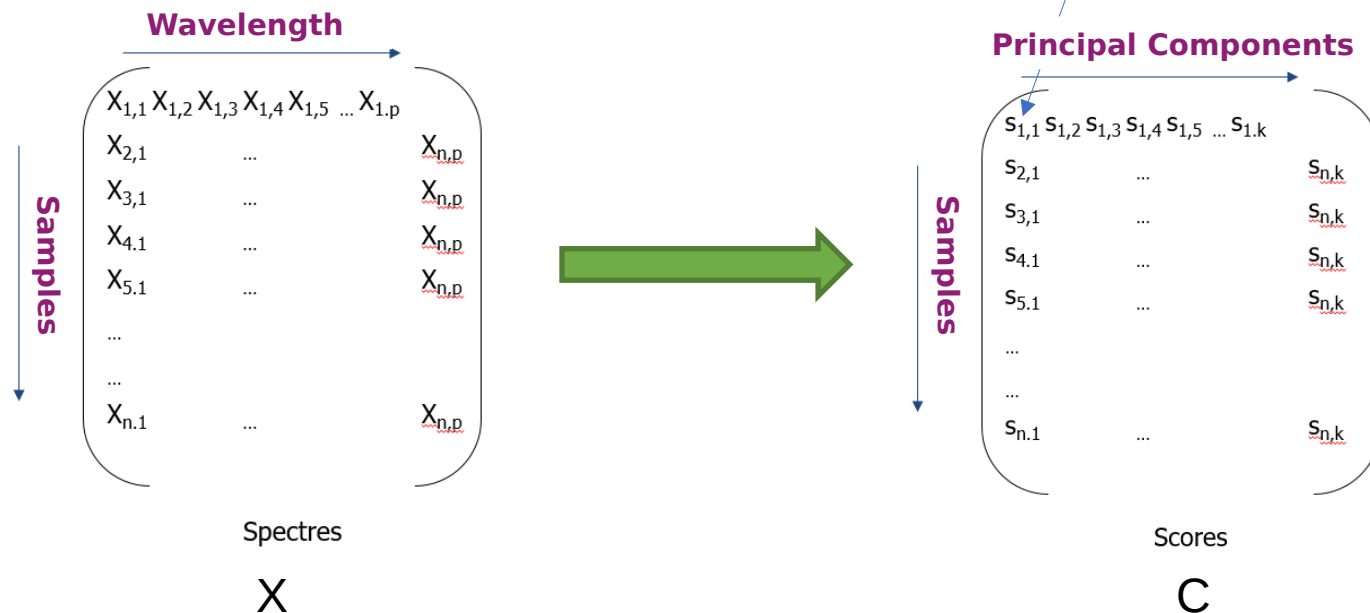


NIRS: How to process data

Model the matrix X with :

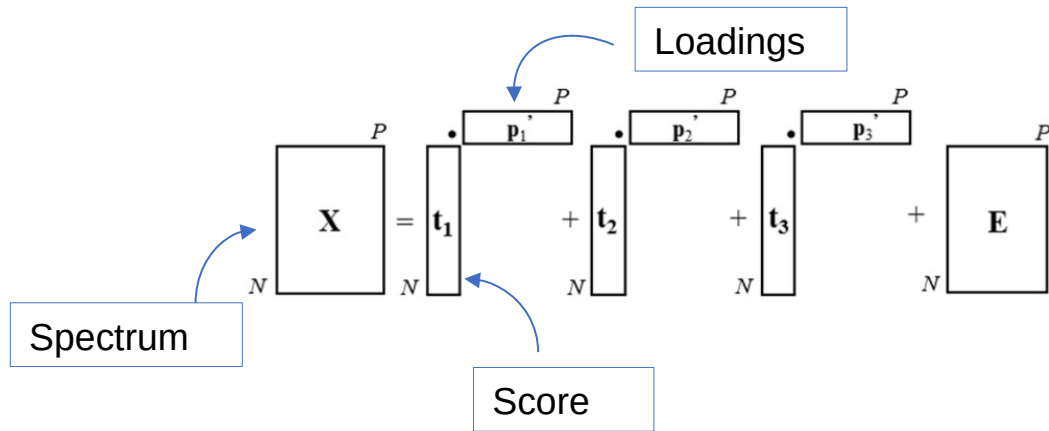
- loadings V (=weight of variables in the model)
- scores C (= coordinates of individuals)

$$X = C V^T + E \quad (E = \text{residuals})$$



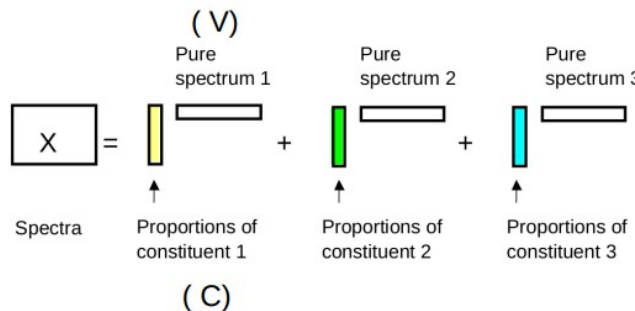
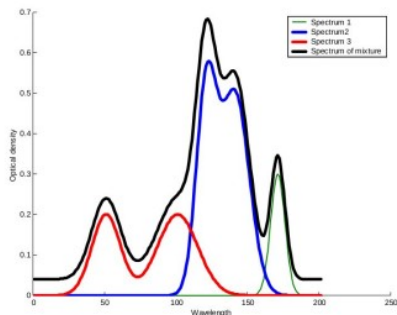
NIRS: How to process data

Principal Component Analysis



NIRS: How to process data

Principal Component Analysis



$$CP1 = \alpha \times \lambda_1 + \beta \times \lambda_2 + \dots + \gamma \times \lambda_p$$

We can evaluate the most important wavelengths for the model.



NIRS: How to process data

Rstudio

- Today: Visualization, Preprocessing, and PCA
- After: Machine Learning